Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine (EALSE)
Maximum Marks	1 Mark

Scalability & Future Plan

This document explains how the Emotion-Aware Learning Support Engine (EALSE) can be expanded to support a larger number of learners, improved performance, cloud deployment, enhanced security, and additional Artificial Intelligence features in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Single-session Streamlit application.	Cannot support many concurrent students or classrooms at once.	High
2	Local CSV-based interaction logging.	Limited scalability; no centralized, multi-user analytics.	High
3	English-only NLTK preprocessing pipeline.	Cannot accurately classify non-English student input.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-session Streamlit application.	Deploy on cloud with multi-user session support and load balancing.
2	Data Storage	Local CSV file (data/interactions.csv).	Use PostgreSQL or Firebase for centralized, queryable storage.
3	Performance	CPU-based dual-model inference (~3.5s end-to-end).	Optimize inference with quantization and enable GPU acceleration.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	API key stored locally in a .env file.	Add user authentication, encrypted storage, and cloud secrets management.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Domain adaptation fine-tuning using real student messages.	1 Month	Improved real-world classification accuracy.
Phase 2	Multi-language preprocessing (Spanish, French, Hindi).	2 Months	Wider accessibility for non-English learners.
Phase 3	Teacher analytics portal aggregating class-wide emotion trends.	4 Months	Actionable, class-wide insights for educators.
Phase 4	Mobile-responsive UI and voice input support.	6 Months	Increased accessibility and wider user adoption.